SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

EXP:7

Name: Balaguru S

Register Number: 192312070

Requirement Analysis and Negative Test Scenarios and Test Cases for a Food Delivery Application

Aim

To study the requirements of a Food Delivery Application, and to prepare negative test scenarios and test cases that confirm the application handles invalid inputs and wrong user actions safely.

Problem Statement

A food delivery application lets customers choose a restaurant, order food, pay online and follow the delivery on a map. It depends on location services, payments and real-time updates, so a defect can mean cold or wrong food, wrong charges or failed orders.

Testing is needed to check that the application:

- Customers can log in, order food and track the delivery correctly.
- Invalid inputs and failures are handled safely.
- Location, cart, payment and cancellation rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The user can register and log in with a mobile number and a one-time password (OTP).
FR2	The app detects the user's location or accepts a manual delivery address inside the service area.
FR3	The user can browse restaurants and filter by cuisine, rating, veg only and delivery time.
FR4	The restaurant menu shows item names, prices, veg or non-veg marks and customization options.
FR5	The user can add items to the cart, change quantities and customise items. Unavailable items cannot be added.
FR6	The user can apply a valid coupon. A coupon works only above its minimum order value.
FR7	Payment is possible by UPI, card, wallet or cash on delivery.
FR8	After payment, the app confirms the order and shows an estimated delivery time.
FR9	The user can track the order status and the delivery partner's live location.
FR10	An order can be cancelled only before the restaurant starts preparing it.
FR11	After delivery, the user can rate the restaurant and the delivery partner.
FR12	A cancelled or failed paid order is refunded to the original payment method.

ID	Requirement
FR13	The user can view past orders and reorder from the history.

2 Software and environment requirements

Item	Details
Application	Android and iOS mobile app, with a web version
Back end and APIs	Java/Node.js services, Google Maps API, SMS gateway for OTP
Database	MySQL / MongoDB (users, restaurants, menus, orders)
Operating system	Android 12+ and iOS 16+ (test devices or emulators)
Testing tools	Test case document (Excel), Appium, Postman, bug tracker (Jira)
Test data	Test users, restaurants with menus, coupon codes, delivery addresses inside and outside the service area, sandbox payment details

Negative Test Scenarios

A negative test scenario checks that the system rejects invalid input or a wrong action, shows a clear message and stays stable and secure.

Scenario ID	Module	Test Scenario	Requirement
FD-NS01	Login	Enter an invalid mobile number	FR1
FD-NS02	Login	Enter a wrong OTP	FR1
FD-NS03	Login	Use an expired OTP	FR1
FD-NS04	Location	Set an address outside the service area	FR2
FD-NS05	Location	Deny location permission and leave the address blank	FR2
FD-NS06	Browse	Search for a restaurant that does not exist	FR3
FD-NS07	Restaurant	Order from a closed restaurant	FR3, FR5
FD-NS08	Cart	Add an unavailable item	FR5
FD-NS09	Cart	Set an invalid quantity	FR5
FD-NS10	Offers	Apply an invalid coupon or one below the minimum order value	FR6
FD-NS11	Payment	Payment fails because of a wrong UPI PIN or insufficient balance	FR7
FD-NS12	Payment	Network drops during payment	FR7, FR12
FD-NS13	Cancellation	Cancel after the restaurant starts preparing	FR10
FD-NS14	Order	Place an order with an empty cart	FR5

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
FD-NT01	FD-NS01	1. Enter a 9-digit mobile number 2. Tap Get OTP	987654321	Error “Enter a valid mobile number” is shown and no OTP is sent.
FD-NT02	FD-NS02	1. Request an OTP 2. Enter an incorrect OTP 3. Tap Verify	123456 (wrong)	Verification fails with “Invalid OTP”. The user is not logged in.
FD-NT03	FD-NS03	1. Request an OTP 2. Wait until it expires 3. Enter the old OTP	OTP older than 5 minutes	“OTP expired” is shown, and the user can request a new OTP.
FD-NT04	FD-NS04	1. Enter an address far from any restaurant 2. Confirm	Address in another city	App shows “We don’t deliver here yet” and does not list restaurants.
FD-NT05	FD-NS05	1. Deny the location permission 2. Leave the address field blank 3. Continue	Permission denied	App asks the user to enter an address manually and does not proceed.
FD-NT06	FD-NS06	1. Type an unknown restaurant name in the search box	Zzzz Biryani House	“No restaurants found” is displayed, and the app does not crash.
FD-NT07	FD-NS07	1. Open a restaurant outside its opening hours 2. Try to add an item	Closed restaurant	Restaurant is marked closed and items cannot be added.
FD-NT08	FD-NS08	1. Open a sold-out item 2. Tap Add	Sold-out item	Add is disabled with an “Item unavailable” message.
FD-NT09	FD-NS09	1. Open the cart 2. Enter 0, then a very large quantity	0; 100	Quantity 0 removes the item, and a limit message is shown for the large quantity.
FD-NT10	FD-NS10	1. Add items worth ₹200 2. Enter FIRST50 3. Then enter a wrong code	FIRST50 (needs ₹300); ABCD99	Both attempts are rejected with a clear reason, and the bill is unchanged.
FD-NT11	FD-NS11	1. Choose UPI 2. Enter a failing test UPI ID 3. Approve the payment	failure@upi	Payment fails, the order is not placed and the user can retry.
FD-NT12	FD-NS12	1. Start the payment 2. Switch off the network before confirmation	Mobile data off	No duplicate order or charge occurs. Any deducted amount is refunded automatically.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
FD-NT13	FD-NS13	1. Open an order marked Preparing 2. Tap Cancel	Order in Preparing status	Cancellation is refused with a message explaining why.
FD-NT14	FD-NS14	1. Open the cart with no items 2. Try to proceed to payment	Empty cart	Proceed button is disabled or the app shows “Your cart is empty”.

Result / Conclusion

Result: 14 negative test scenarios and 14 test cases were prepared for the Food Delivery Application, covering 8 of the 13 functional requirements (no negative scenario for FR4, FR8, FR9, FR11, FR13). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	3	3
Location	2	2
Browse	1	1
Restaurant	1	1
Cart	2	2
Offers	1	1
Payment	2	2
Cancellation	1	1
Order	1	1
Total	14	14

Conclusion: The negative test cases cover invalid inputs, boundary violations, failed payments or operations, and unauthorized actions in the Food Delivery Application. When they are executed on a build, they confirm that the system rejects wrong input safely, shows clear messages, and keeps the data and the user’s session secure. Together with the positive test cases, they give a complete set of tests for the stated requirements.